Date

CPU Clock Cycles (CC) = No. of total Instruction (IC) * Average No. of Cycle per Instruction (CPI)

LECTURE-04

Monday 31/8/26

☐ Instruction Count & CPI.

CPU Time = Instruction Count (IC) x CPI x Clock Cycle Time

$$\text{CPU Time} = \frac{IC \times CPI}{\text{Clock Rate}}$$

Q: A: Cycle time = 250 ps, CPI = 2
B: " = 500 ps, CPI = 1.2 } Same ISA

Which is faster & how much?

CPU Time A = IC x CPI_A x Cycle time (A)
= I x 2 x 250 = $\boxed{500\,ps}$

CPU Time B = IC x CPI_B x Cycle time (B)
= I x 1.2 x 500 = $\boxed{600\,ps}$

$$\frac{\text{Time}(B)}{\text{Time}(A)} = \frac{600\,I}{500\,I} = \boxed{1.2}$$ → B is 1.2 times faster.

Q:

Program	CPI
A	1
B	2
C	3

$$\sum_{i=1}^{n} (IC_i \times CPI_i)$$

Date

Class	A	B	C
CPI	1	2	3
Seq 1	2	1	2
Seq 2	4	1	1

IC

Seq 1 & 2 tells kay specific sequence main kitne total Instruc (IC) hein

• CPU Cycles Seq 1 = $(1\times2)+(2\times1)+(3\times2)$ = $\boxed{10 \text{ Cycles}}$

total in seq 1

• $IC_{(1)} = 2+1+2 = \boxed{5}$

~~The Seq Sequence 2~~

Seq 2 • CPU Cycles = $(1\times4)+(2\times1)+3\times1$ = $\boxed{9 \text{ Cycles}}$

total in Sequence 2.

• $IC_{(2)} = 4+1+1 = \boxed{6}$

• Therefore Sequence two is better as it has less cycles (9) than seq. 1 eventhough IC of seq 2 is greater than seq 1 IC

RISC-V : It is a computer which uses reduced instruction set.

$y = (g+h) - (i+j)$

$g+h$ → t0, $i+j$ → t1

add t0, g, h
add t1, i, j
Sub y, t0, t1.

Double = 64 bits
Word = 32 bits

① R-Type: it's a type of RISC-V

• It has 32 registers (so we need 5 bits as $2^5 = 32$)

• Each register has 64 bits.

add x5, x20, x21
add x6, x22, x23
sub x19, x5, x6.

these are R-type Instruction.

• replaced the variables with registers (like : x21) • we can't use x0.

• need to tell that we're assigning which variable to which registe.

offset to add
to get index of address

Load : Memory → register (read)

ld , x5 , address ⟶ 40 (x28) Base address

double

Store : Register → Memory (write)

sw , x5 , address

word

* It could be ld, lw, sd, sw.

Sign extension:

- Representing a number using more bits to preserve the numeric value
- Whichever bit will be in ~~MSB~~ MSB we just have to extend it towards left.
- if we've 1011 in signed form then it is $\boxed{-3}$ we'll just do 1111 1011 for sign ext.

⇒ Instruction Register & Bits Division:

7 bits	5 bits	5 bits	3 bits	5 bits	7 bits
func7	rs2	rs1	func3	rd	opcode

add x9, x20, x21

0000000	10101	10100	000	01001	0110011

$(0\ 1\ 5\ A\ 0\ 4\ B\ 3)_{16}$

func3 = 0

func7 = 0

opcode = 51

all these and the registers should be in binary of specific boxes's bit length.